

EXP NO 6	CDN SERVICE SELECTION AND CONFIGURATION USING CLOUDFLARE
Date: 21.08.2026	

Aim

To select a Content Delivery Network (CDN) provider, deploy a simple web application using Cloudflare Workers, access the application through Cloudflare's edge network, and verify the HTTP response and Cloudflare-specific headers using browser Developer Tools.

Requirements

- Windows 10/11
- Google Chrome or any modern web browser
- Cloudflare Account
- Internet Connection
- Chrome Developer Tools

Procedure

Part A – Select the CDN Provider

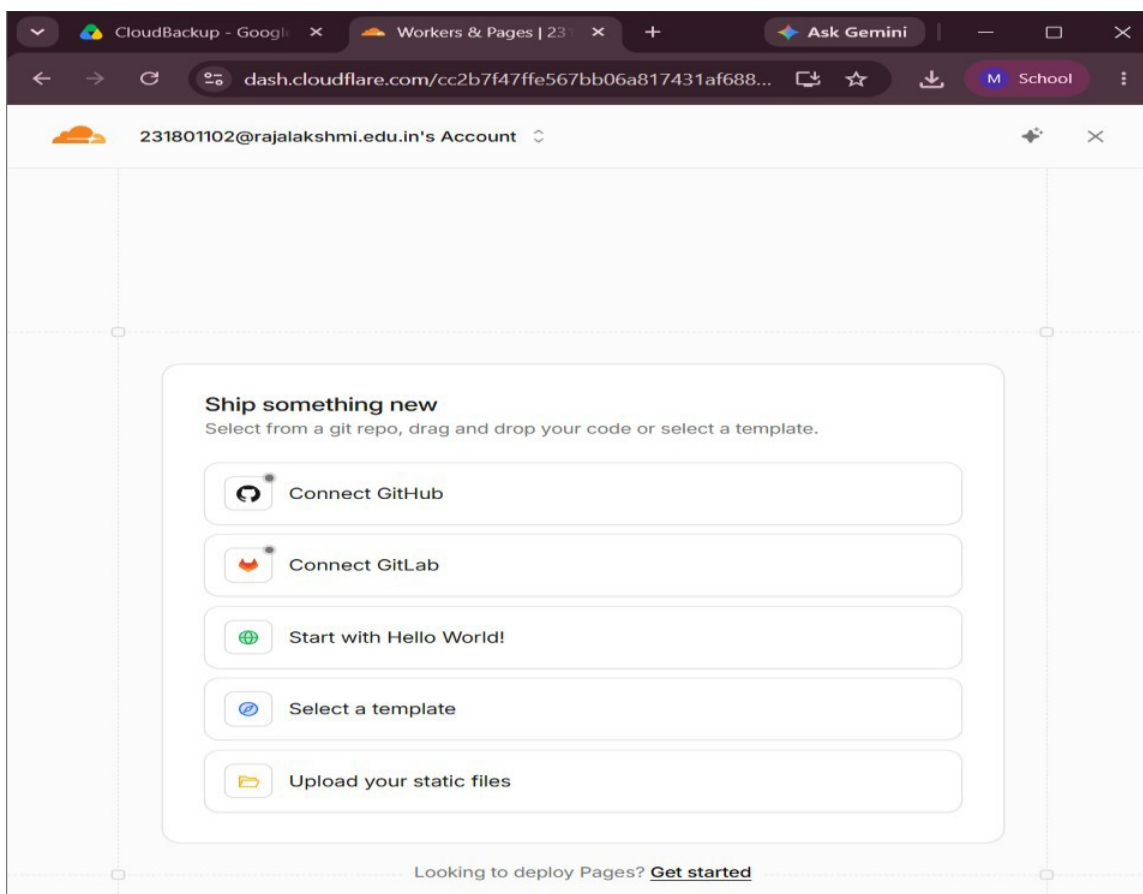
Step 1: Select a CDN Provider

A Content Delivery Network provider was selected for implementing the experiment. The available CDN providers considered included Cloudflare, Amazon CloudFront, and Akamai. Cloudflare was selected because it provides CDN, security, DNS, and edge computing services, and allows simple deployment of applications using Cloudflare Workers.

Part B – Create and Deploy a Cloudflare Worker

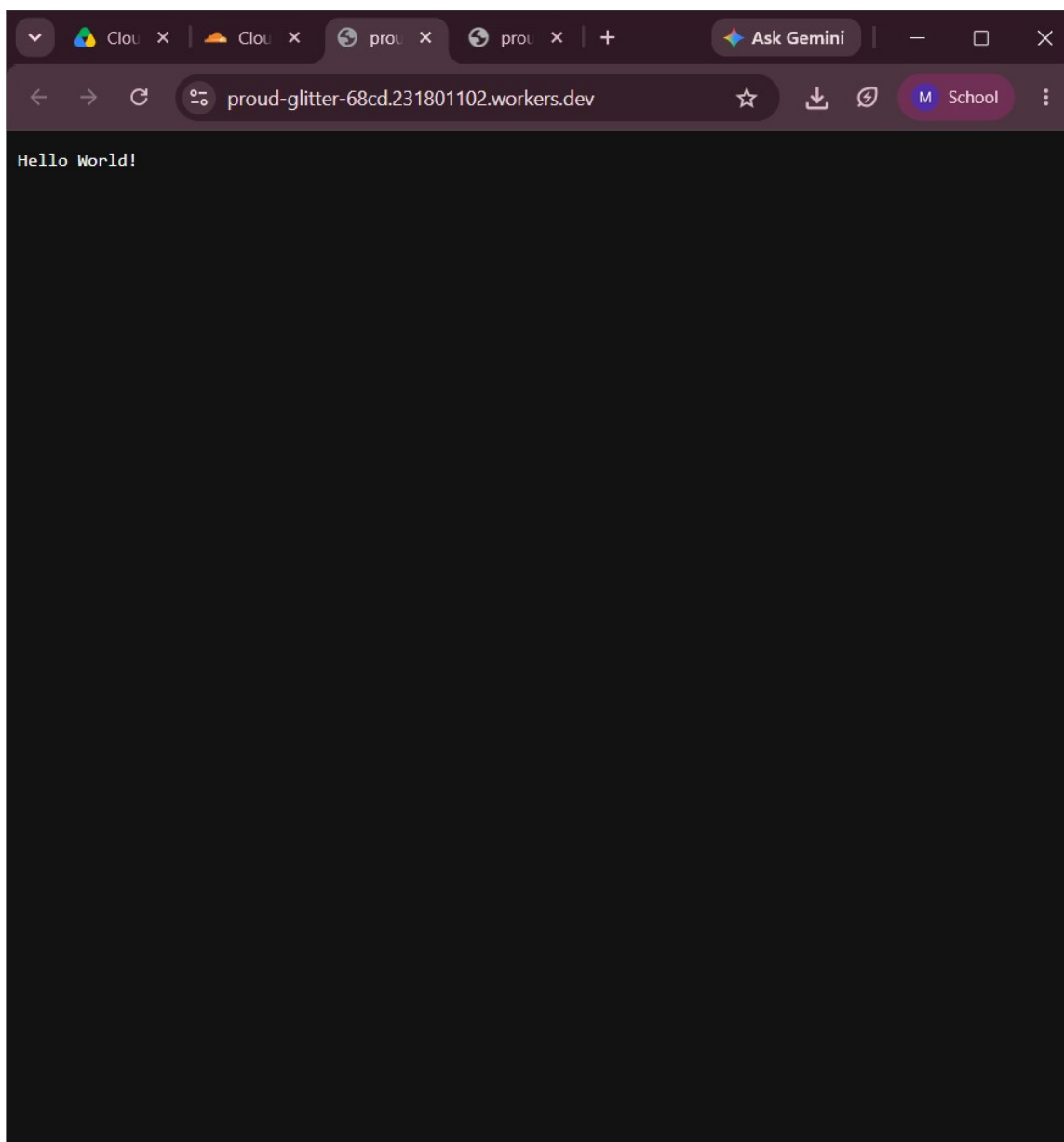
Step 2: Access Cloudflare

The Cloudflare dashboard was accessed through a web browser by logging in with a valid Cloudflare account.



Step 3: Create a New Application

The Workers & Pages service was opened from the Cloudflare dashboard. The Create App option was selected, followed by Start with Hello World!, which created a simple Cloudflare Worker application.



Step 4: Deploy the Hello World Application

The default Worker code was retained and deployed. The application returned the response “Hello World!”, and Cloudflare generated a public URL ending with workers.dev. The deployed application was accessed successfully through the generated public URL.

Part C – Create the CDN Demonstration Webpage

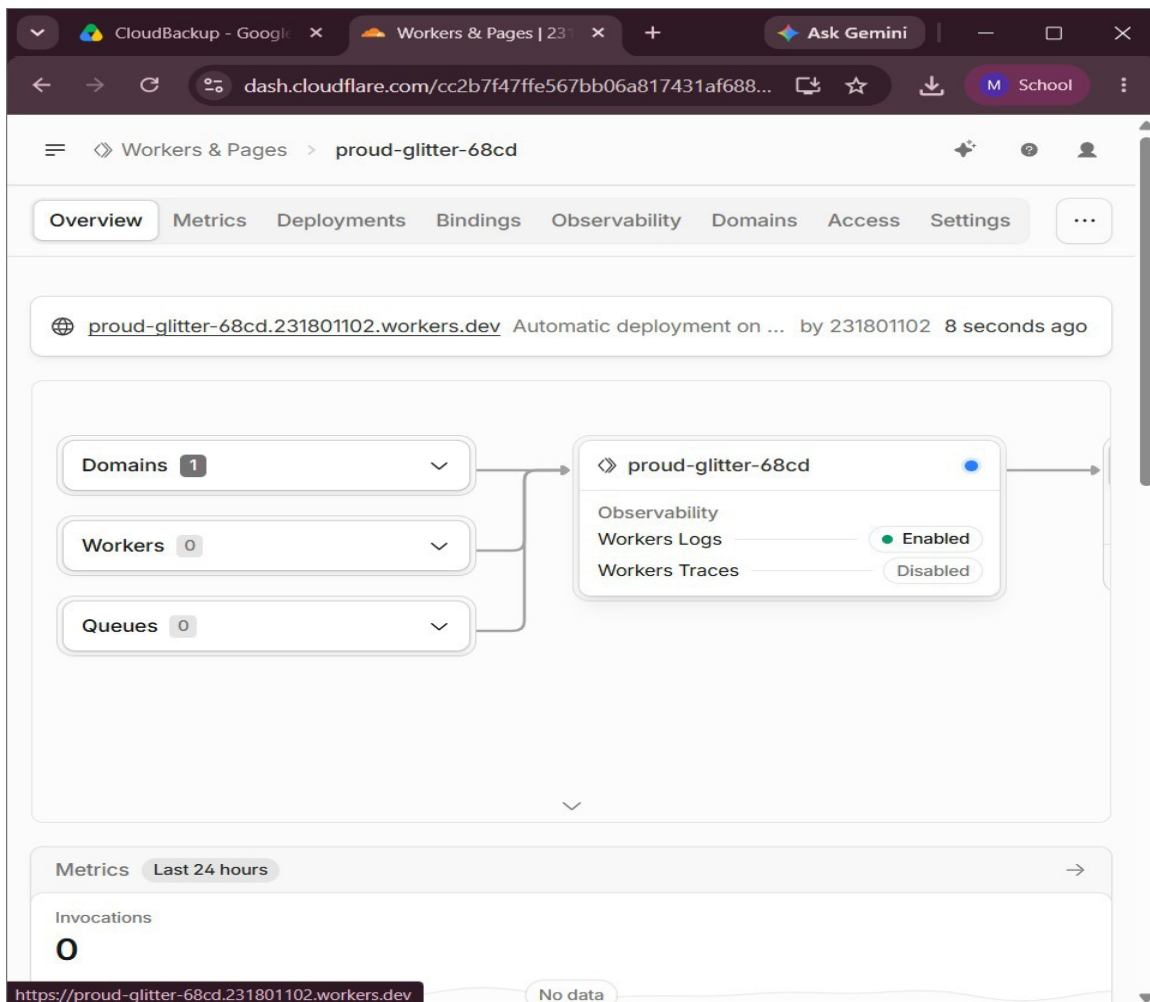
Step 5: Edit the Worker Code

The Cloudflare Worker code editor was opened, and the default Hello World response was replaced with a simple HTML webpage containing information about CDN content delivery, including:

- CDN Service Experiment
- Cloudflare Content Delivery
- This webpage is being delivered through Cloudflare
- CDN reduces the distance between users and content
- Status: Successfully Deployed

Step 6: Deploy the Updated Application

After modifying the Worker code, the application was deployed again. The updated webpage was accessed using the public workers.dev URL, and the CDN demonstration webpage was successfully displayed in the browser.



Part D – Verify the HTTP Response**Step 7: Open Browser Developer Tools**

The deployed Cloudflare Worker webpage was kept open in Google Chrome, and Chrome Developer Tools were opened by pressing F12. The Network tab was selected.

Step 8: Reload the Webpage

The webpage was reloaded while the Network tab was active, and the main request generated for the Cloudflare Worker was displayed in the Network activity list.

Step 9: Inspect the Request Headers

The main Worker request was selected and the Headers section was opened. The following information was verified:

- Request Method: GET
- Status Code: 200 OK
- Content-Type: text/html
- Server: cloudflare
- CF-Ray: Cloudflare request identifier

The screenshot shows a web browser window with the address bar displaying `proud-glitter-68cd.231801102.workers.dev`. The page content on the left includes the title "CDN Service Experiment", a subtitle "Cloudflare Content Delivery", and a status "Status: Successfully Deployed". The right side of the browser shows the Network tab with a list of requests. The selected request is for `favicon.ico`, and its details are expanded in the right pane.

CDN Service Experiment

Cloudflare Content Delivery

This webpage is being delivered through Cloudflare.

CDN reduces the distance between users and content.

Status: Successfully Deployed

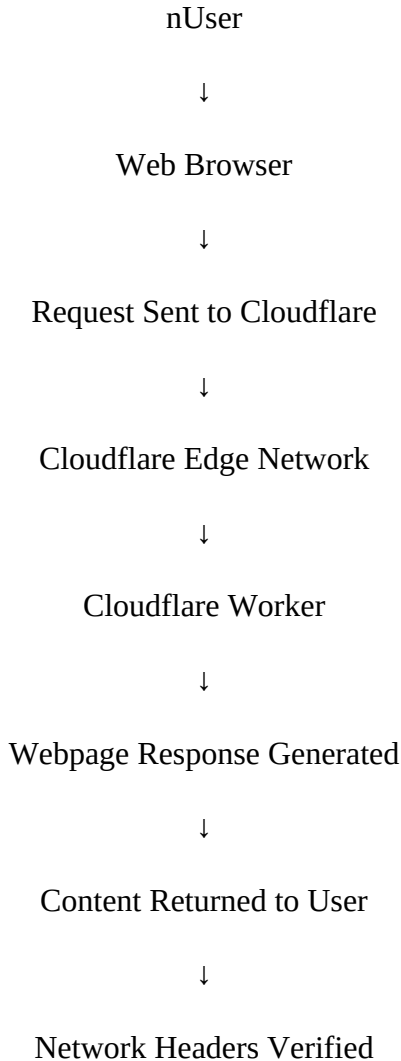
Network Tab Details:

Name	Headers	Preview	Response	Initiator
proud-glitter-68cd.231801102...	General			
favicon.ico	Request Headers: - Request URL: <code>https://proud-glitter-68cd.231801102.workers.dev/</code> - Request Method: <code>GET</code> - Status Code: <code>200 OK</code> - Remote Address: <code>[2606:4700:3033::6815:1b41]:443</code> - Referrer Policy: <code>no-referrer</code> Response Headers: - Alt-Svc: <code>h3=":443"; ma=86400</code> - Cf-Ray: <code>a31cbf3018b76d97-SIN</code> - Content-Encoding: <code>zstd</code> - Content-Type: <code>text/html</code> - Date: <code>Thu, 27 Aug 2026 17:20:25 GMT</code> - Nel: <code>{"report_to":"cf-nel","success_fraction":0.0,"max_age":604800}</code> - Priority: <code>u=0,i</code> - Report-To: <code>{"group":"cf-nel","max_age":604800,"endpoints":[{"url":"https://a.nel.clo</code>			

2 requests | 1.3 kB transferred | 8

The Server: cloudflare header confirmed that the response was handled through Cloudflare's network. The CF-Ray header provided a unique identifier for tracing the request through Cloudflare's infrastructure.

CDN DELIVERY PROCESS



Conclusion

The CDN service selection and configuration experiment was successfully carried out using Cloudflare. A Cloudflare Worker was created and deployed on Cloudflare's distributed edge network. The deployed application was accessed through a public workers.dev URL and successfully delivered the CDN demonstration webpage. Browser Developer Tools were used to inspect the network request; the response returned an HTTP status of 200 OK and included Cloudflare-specific headers such as Server: cloudflare and CF-Ray, confirming that the application request was processed and delivered through Cloudflare's edge network.